

Emergent Gravity Program: Final Status of Claims and Cross-References (14 August 2026)

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Abstract

This document is the authoritative final status of the Emergent Gravity Program. It records every major claim with its current epistemic label, incorporates all cross-references among the papers of the series, and states the remaining open items with precision. The AdS linearized sector (metric and axial) is closed. The high-precision numerical coefficient of the kinematic residue is known only to order of magnitude. The de Sitter domain, the non-linear regime, and UV completion remain open. No stronger claim is made than the evidence supports.

Epistemic Conventions

[P] = proved/derived within the stated framework. [C] = conjectured with nontrivial support.
[O] = open. Numerical agreement is not proof. Consensus is not proof.

1 Final Status of Major Claims

Claim	Label	Notes
FGHNV first-law $\leftrightarrow$ linearized Einstein (AdS)	[P]	External literature
Modular perturbation kernel (exact + machine-verified)	[P]	Paper IV
Metric sector of Condition NL unconditional	[P]	Conformal selection rule
$\mathcal{I}_B$ non-vanishing for finite regions	[P]	Paper VI + numerical campaign
Absorbability of the residue	[P]	Paper IV
Relative residue of $\mathcal{I}_B$ is $\mathcal{O}(1)$	[P]	Version 4 status note
Multi-digit coefficient of $\mathcal{I}_B$	[O]	Requires analytic Hadamard expansion
Sign of Hehl–Datta coefficient	[P]	Relative-entropy positivity
Magnitude by rigidity	[P]	RT + modular selection rule
Magnitude from pure information	[P]	Null-plane limit + absorbability
de Sitter / cosmological domain	[O]	Paper IX
Non-linear regime	[O]	No derivation at FGHNV standard
UV completion	[O]	Open by construction

2 Series Cross-References

- Paper I — Entanglement as the Origin of Spacetime Curvature
- Paper II — Spin-Current Sources and Ball Modular Hamiltonians
- Paper III — The New Standard Model for Unified Physics
- Paper IV (Updated) — Condition NL for Finite Balls
- Paper V (Updated) — The Hehl–Datta Coefficient from Relative-Entropy Positivity
- Paper VI — Evaluation of the Universal Kinematic Integral
- Paper VII — Pure-Information Determination of the Hehl–Datta Coefficient
- Paper VIII (Version 4) — Status of the High-Precision Coefficient of $\mathcal{I}_B$
- Paper IX — Domain of Validity of the Emergent Gravity Program
- Monograph — Emergent Gravity and a New Lagrangian

All later papers cite the earlier ones; the updated IV and V cite VI–IX; the status notes cite the entire series. The bibliography below is the complete reference list for the program as of this date.

3 What Is Closed

Inside the holographic AdS regime with ball-shaped regions:

- the finite-ball torsionful first law holds (metric sector unconditional, axial sector after finite dictionary improvement); - the Hehl–Datta coefficient is fixed in sign by relative entropy and in magnitude both by rigidity and by pure information; - the kinematic residue $\mathcal{I}_B$ is non-vanishing and of relative size $\mathcal{O}(1)$, and is absorbable.

4 What Remains Open

- Multi-digit numerical value of the coefficient of $\mathcal{I}_B$ (analytic Hadamard subtraction still required). - Extension beyond AdS (de Sitter / cosmology). - Non-linear Einstein–Cartan equations from entanglement. - UV completion of the effective Lagrangian.

These limitations are recorded with the same precision as the positive claims. The program is a theory of everything we know inside its stated domain; it is not a theory of everything.

References

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